

Table 1. Comprehensive statistical summary of hybrid effects on immune-predicted weight
Multiple analytical approaches converge on sex-specific, infection-dependent hybrid dysfunction

Analysis	Statistical Test	p-value	Biological Interpretation
1. Complete Dataset (n=304) ¹	Overall hybrid effect	0.017*	Significant overall I
	Males only	0.038*	Males show hybrid
	Females only	0.189	Females show no eff
	Sex-specific (H3 vs H2) ²	0.027*	Sex-specific effects
2. Uninfected Only (n=171)	Overall hybrid effect	0.545	No constitutive cos
	Males only	0.432	No effect in uninfected
	Females only	0.908	No effect in uninfected
3. By Infection Status (n=304)	Infection groups differ (H2 vs H0)	0.002**	Strong infection do
	Uninfected hybrid effect	0.545	No effect without infection
	Infected hybrid effect	0.293	No effect with infection
4. Males Only - Infection (n=153)	Infection groups differ (H1 vs H0)	0.010**	Infection status critical
	Males only	0.016*	Males affected by infection
5. Infected Only (n=133)	Sex-specific (H3 vs H2) ²	0.027*	Sex-specific in infected
	Males only	0.483	No effect in infected
6. Linear Regression	Sex × HI interaction	0.011*	Independent confirmation

¹ parasiteloading framework tests deviation from additive subspecies expectations

² Critical test for sex-specific effects using advanced models

Significance: **p < 0.01, *p < 0.05. Green highlighting indicates significant effects supporting hybrid dysfunction.

Key finding: Males show consistent hybrid dysfunction across analyses, manifesting only during infection.

